

5.1

0,0,0,0; 0,0,0,1; 0,0,0,2; 0,0,1,0; 0,0,1,1; 0,0,1,2; 0,0,2,0; 0,0,2,1; 0,0,2,2;
 0,1,0,0; 0,1,0,1; 0,1,0,2; 0,1,1,0; 0,1,1,1; 0,1,1,2; 0,1,2,0; 0,1,2,1; 0,1,2,2;
 1,0,0,0; 1,0,0,1; 1,0,0,2; 1,0,1,0; 1,0,1,1; 1,0,1,2; 1,0,2,0; 1,0,2,1; 1,0,2,2;
 1,1,0,0; 1,1,0,1; 1,1,0,2; 1,1,1,0; 1,1,1,1; 1,1,1,2; 1,1,2,0; 1,1,2,1; 1,1,2,2;

5.2

$$(1) 6 \times 6 \times 8 = 288$$

$$(2) 1000 + 288 - 6 = 1282$$

$$(3) 1000 + (2 \times 8 + 4) \times 6 = 1120$$

$$(4) 1000 + (4 \times 6 + 2) \times 6 = 1156$$

5.3

$$\begin{aligned} k &= n + (n - 1) + \dots + (n - i + 2) + (j - i + 1) - 1 \\ &= -\frac{1}{2}i^2 + (n + \frac{1}{2})i + j - (n + 1) \\ \Rightarrow f_1(i) &= -\frac{1}{2}i^2 + (n + \frac{1}{2})i ; f_2(j) = j ; c = -(n + 1) \end{aligned}$$

5.4

$$\begin{aligned} k &= 4 \times \left\lfloor \frac{i-1}{2} \right\rfloor + 2 \times ((i-1)(mod2)) + (j-1)(mod2) \\ &= 2(i-1) + (j-1)(mod2) \end{aligned}$$

5.5

三元组 i j a_{ij}

0	1	1
0	4	5
1	0	2
1	1	3
1	3	6
3	1	4
3	4	7

十字链表. $rhead[0] \rightarrow (0, 1, 1) \xrightarrow{\text{right}} (0, 4, 5) \rightarrow \text{NULL}$

$rhead[1] \rightarrow (1, 0, 2) \rightarrow (1, 1, 3) \rightarrow (1, 3, 6) \rightarrow \text{NULL}$

$rhead[2] \rightarrow \text{NULL}$

$rhead[3] \rightarrow (3, 1, 4) \rightarrow (3, 4, 7) \rightarrow \text{NULL}$

$chead[0] \rightarrow (1, 0, 2) \xrightarrow{\text{down}} \text{NULL}$

$chead[1] \rightarrow (0, 1, 1) \rightarrow (1, 1, 3) \rightarrow (3, 1, 4) \rightarrow \text{NULL}$

$chead[2] \rightarrow \text{NULL}$

$chead[3] \rightarrow (1, 3, 6) \rightarrow \text{NULL}$

$chead[4] \rightarrow (0, 4, 5) \rightarrow (3, 4, 7) \rightarrow \text{NULL}$